

Standalone SMITH / SONIC manual

SMITH/SONIC standalone distribution

1. Scope

Standalone SMITH converts a Cartesian molecular description and its perceived molecular state into a rank-complete SONIC internal-coordinate contract. It is the self-contained, reproducible distribution associated with the SMITH manuscript.

The standalone input boundary is provider-neutral: SMITH accepts supplied topology, supplied redundant primitive/Wilson-B definitions, a complete frozen state, or plain Cartesian geometry through a bundled perception frontend. The bundled frontend is a revision-pinned subset of ORACLE and constructs ordinary topology, molecular point-group operations, atom permutations and primitive coordinates. It is included so that standalone SMITH has no runtime dependency on another repository.

An input provider may supply topology, cycles, symmetry, atom equivalence, continuous descriptors, interaction centres, and the redundant primitive/Wilson-B source. SMITH constructs coordinate candidates, protected special coordinates, rank reduction, symmetry adaptation, analytic Wilson rows, and the serialized SONIC contract. Its boundary ends there. Optimization, scans, finite internal-to-Cartesian realization, force fields, Hessian transport and higher derivatives belong to applications consuming the contract. In particular, B-prime is not required, constructed, or stored by SMITH; it is evaluated on demand by a second-order consumer when needed.

The corresponding MATRIX ownership is: ORACLE provides authoritative symmetry, primitive coordinates and B; SMITH consumes them without re-perception and builds SONIC; LINK realizes internal-coordinate changes in Cartesian space; ARCHITECT evaluates B-prime and transforms nonstationary Hessians. MORPHEUS delegates realization to LINK, while SENTINEL proposes only SONIC points through the LINK protocol.

2. Requirements and installation

Use Python 3.11 or newer and Git. The release candidate contains all required SMITH sources; no second repository or machine-specific path is needed.

```
gh auth login
gh auth setup-git
export SMITH_CHECKOUT=/path/to/your/Smith
export SMITH_ENV=/path/to/your/smith-venv
git clone https://github.com/yogibubu/Smith.git "$SMITH_CHECKOUT"
cd "$SMITH_CHECKOUT"
git switch --detach v0.1.0rc7
python3 -m venv "$SMITH_ENV"
source "$SMITH_ENV/bin/activate"
python -m pip install --upgrade pip
python -m pip install ./standalone
```

Replace the two `/path/to/your/...` values with directories chosen by the user. On Windows PowerShell, define the corresponding paths and activate the environment with `<smith-env>\Scripts\Activate.ps1`. The detached checkout is intentional and fixes the complete standalone source snapshot.

3. Command summary

```
smith-sonic --version
smith-sonic example NAME [OUTPUT]
smith-sonic build INPUT [OUTPUT] [--require-frozen-state]
smith-sonic inspect FILE
```

`example` runs an input installed with the package. `build` accepts a plain extended XYZ input, an XYZ carrying TOPOLOGY or PRIMITIVES, or a complete frozen xyzin. `inspect` reports the presence of the frozen state, GIC section, and provenance profile.

Every `build` or `example` command writes three files with the same stem:

- `.xyzin`: frozen SMITH/SONIC contract;
- `.smith.out`: readable coordinate report with rank, families, protected rows, values, units, irreps, active/frozen status, and primitive coefficients;
- `.g16.gjf`: Gaussian 16 Rev. C.01 `ReadAllGIC` optimization input.

Gaussian 16 Rev. C.01 is used because its general `ReadAllGIC` language can interpret SONIC expressions directly and independently. SONIC serialization is one particular use of this much broader GIC language. The profile is enabled by default, uses Gaussian-compatible improper dihedrals, and marks every non-totally symmetric coordinate as `Frozen`; totally symmetric coordinates remain active. Fragment and interaction-centre helper functions are serialized as `Inactive` definitions. Gaussian 16 Rev. C.01 does not support native SONIC out-of-plane rows or every special/multi-periodic composite expression. The `.g16.gjf` writer therefore performs a terminal compatibility translation; native out-of-plane and special coordinates remain unchanged in `.xyzin` and in the human-readable report.

For a complete externally prepared and validated state, require the boundary explicitly:

```
smith-sonic build molecule.xyzin molecule.sonic.xyzin --require-frozen-state
```

This option refuses an input missing any of `VALIDATION`, `TOPOLOGY`, `SYNTHONS`, `SYMMETRY`, or `PRIMITIVES`. The latter contains the ordered redundant coordinates, reference values and Wilson-B fingerprint.

4. Input profiles

Complete frozen state

This is the production interface. SMITH preserves the molecular perception and primitive/B sections and adds the SONIC construction. The output provenance contains `PERCEPTION_PROFILE FROZEN_STATE`.

Supplied topology or primitives

An input carrying `TOPOLOGY` records `STANDALONE_TOPOLOGY` and generates the ordinary redundant primitive set from that graph. An input carrying `PRIMITIVES` records `STANDALONE_PRIMITIVES` and validates the supplied reference values and Wilson-B fingerprint. If both sections are present, their covalent bond rows must agree.

Minimal Cartesian profile

A plain or extended XYZ is sufficient for small reproducibility examples. The packaged path performs the embedded topology/symmetry/primitive pass and records `PERCEPTION_PROFILE STANDALONE_MINIMAL`; the profile name is retained for schema compatibility. For disconnected components it also constructs fragment definitions and six intermolecular translation/orientation coordinates. This profile is deliberately limited; advanced continuous descriptors, nondefault quasi-symmetry choices, externally audited symmetry and interaction centers must be supplied in the input when they affect the requested SONIC contract.

Extended-XYZ directives follow the Cartesian block, for example:

```
#SMITH
fragment_mode = special-coordinates
symmetrize = false
sycart = false
```

```
improper_dihedrals = false
```

The available advanced controls include `symmetry_group`, `xh_stretch_policy`, `local_xh_bonds`, and `local_xh_classes`. Omit them to use the pinned SMITH defaults.

5. Packaged examples

Run the complete set with:

```
smith-sonic example water water.xyzin
smith-sonic example norbornane norbornane.xyzin
smith-sonic example formic-acid-water formic-acid-water.xyzin
smith-sonic example water-dimer water-dimer.xyzin
smith-sonic example benzene-water benzene-water.xyzin
smith-sonic example eta3-allyl-palladium eta3-allyl-palladium.xyzin
```

Expected results are:

Example	Input profile	Target rank	Purpose
water	standalone minimal	3	minimal nonlinear molecule
norbornane	standalone minimal	51	bridged and cyclic topology
formic-acid-water	standalone minimal + fragments	18	non-covalent two-fragment contract
water-dimer	standalone minimal + fragments	12	hydrogen-bonded two-fragment contract
benzene-water	standalone minimal + fragments	39	aromatic-polar two-fragment contract
eta3-allyl-palladium	complete frozen state	24	protected metal-to- η^3 -centre coordinate

Each of the three non-covalent outputs must contain three `FRAG_TRANSLATION` and three `FRAG_ORIENTATION` primitives. The η^3 probe must retain an `ETA3_CENTER` over the three allyl carbon atoms and create a protected `CENTER_ATOM_DISTANCE` from Pd to that centre.

The η^3 geometry is an idealized interface test, not an optimized or computed chemical benchmark. Its interaction centre is supplied explicitly in a frozen input fixture. It tests whether SMITH consumes the correct contract; it does not claim that the minimal profile can perceive η^3 bonding.

The advanced generated artifacts are retained under `standalone/examples/`:

- `formic_acid_water.xyzin`, `.smith.out`, and `.g16.gjf`;
- `water_dimer.smith.xyz` and `benzene_water.smith.xyz`;
- `eta3_allyl_palladium.xyzin`, `.smith.out`, and `.g16.gjf`.

6. Reading an output

The most relevant sections are:

- `TOPOLOGY`, `SYNTHONS`, `SYMMETRY`, and `INTERACTION_CENTERS`: perceived molecular state consumed by SMITH;
- `GIC`: build options, target dimension, primitive candidates, protected special coordinates, and the selected generalized internal coordinates;
- `SMITH_PROVENANCE`: package version, implementation revision, and input and fragment profiles.

For a nonlinear system of N atoms without disconnected fragments, the usual target is $3N - 6$. Always use the serialized `TARGET_RANK` and reported final rank when special fragment or centre coordinates are present.

7. Verification and problem reports

For the complete standalone publication verification, run:

```
python -m pip install "./standalone[test]"
python -m pytest standalone/tests/test_examples.py
```

The public gate checks all input profiles, both output formats, the three non-covalent examples, and the supplied interaction-centre fixture. Cross-application integration tests remain development checks and are not installation requirements for standalone SMITH.

The LINK/MORPHEUS integration probe is intentionally separate because those applications are not standalone dependencies. Developers can run it against a compatible clean MATRIX checkout with:

```
MATRIX_ROOT=/absolute/path/to/MATRIX SMITH_REQUIRE_DOWNSTREAM=1 \
python -m pytest standalone/tests/test_downstream_consumers.py
```

When reporting a problem, include the operating system and architecture, Python version, exact command, complete terminal output, input file, and generated `xyz.in` if one was produced. Do not include credentials or private tokens. Open an issue in the repository or send the report directly to the project owner, and identify the frozen release version being tested.

Execution of Gaussian jobs and all optimization or finite-displacement workflows remain outside this package. SMITH generates the Gaussian 16 input but does not launch the external executable.